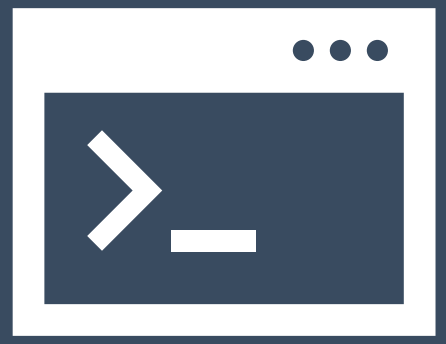
Current Topics in Bioinformatics

https://hbctraining.github.io/Training-modules/



Harvard Chan Bioinformatics Core

HBC training team: hbctraining@hsph.harvard.edu
HBC consulting: bioinformatics@hsph.harvard.edu

HARVARD MEDICAL SCHOOL

DF/HCC DANA-FARBER / HARVARD CANCER CENTER

CFAR HARVARD UNIVERSITY CENTER FOR AIDS RESEARCH

HARVARD T.H. CHAN SCHOOL OF PUBLIC HEALTH

Training

- **Basic Data Skills**
  - Shell
  - R
- **Advanced Topics: Analysis of high-throughput sequencing data**
  - Chromatin Biology
  - Bulk RNA-seq
  - Differential Gene Expression
  - scRNA-seq
  - Variant Calling
- **Current Topics in Bioinformatics**

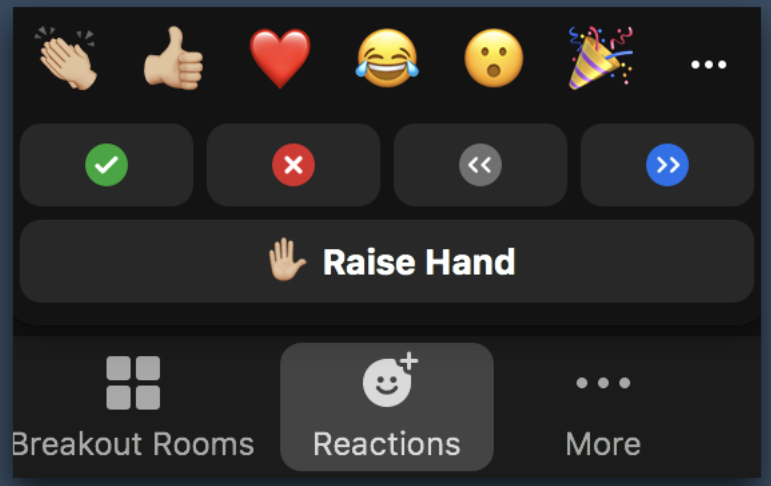
Consulting

- **Transcriptomics:** RNA-seq, small RNA-seq, scRNA-seq
- **Epigenetics:** ChIP-seq, genome wide methylation, ATAC-seq
- **DNA Variation:** WGS, resequencing, exome-seq, CNV studies
- **Functional enrichment analysis**
- **Experimental design help**
- **Grant support**

https://bioinformatics.sph.harvard.edu/

Odds & Ends

- Quit/minimize all applications that are not required for class
- Are you all set?
  - = "agree", "I'm all set"
  - = "disagree", "I need help"



Odds & Ends

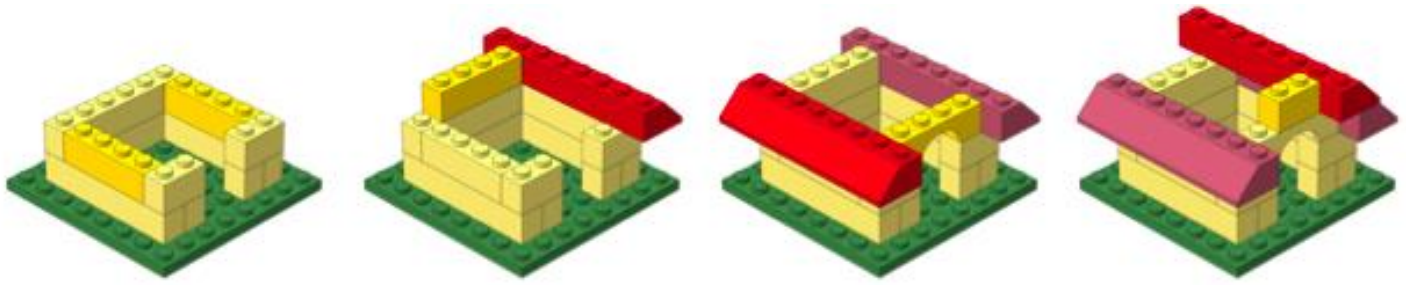
- Questions for the presenter?
  - Post the question in the Chat window OR
  -

when the presenter asks for questions
  - Let the Moderator know
- Technical difficulties with software?
  - Start a private chat with the Moderator with a description of the problem

Exit survey

http://tinyurl.com/hbc-modules

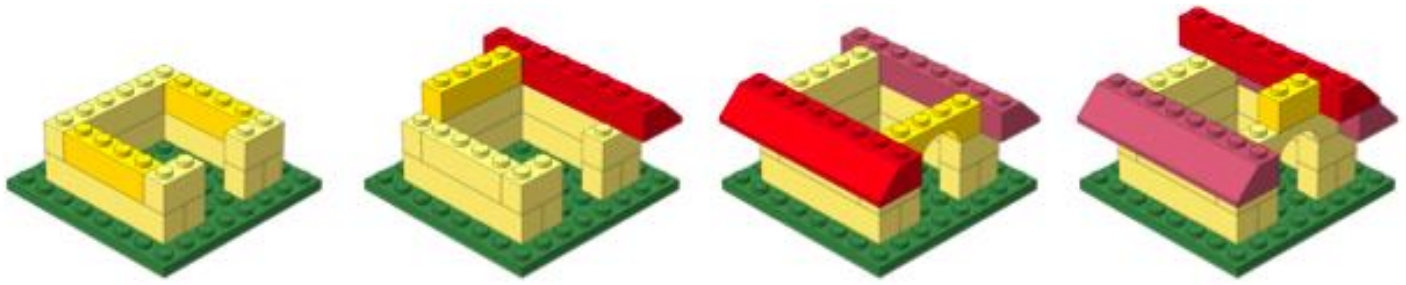
Keep building!



Module	Pre-requisite	Date	Time	Registration Page
"Track Changes" with your code: An Introduction to Git and GitHub	Foundations in R	4/15/26	9:30am-12:30pm	Register here
A Practical Introduction to Nextflow and nf-core	None	5/20/26	9:30am-12:30pm	Register here
Getting Started with Nextflow Workflow Development	Basic experience with command line and scripting concepts (e.g. bash, python, R or equivalent)	6/10/26	9:30am-12:30pm	Register here

https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/

Keep building!



Workshop	Dates	Time	Location	Registration Page*
Introduction to Python	April 14, 17, 21 & 24, 2026	9:30am-12pm	Zoom	TBA
Introduction to Python	July 7, 10, 14 & 17 2026	9:30am-12pm	Zoom	TBA
Introduction to R	August 28, September 1, 4 & 8 2026	10am-12pm	Zoom	TBA
Introduction to Python	November 10, 13, 17 & 20 2026	9:30am-12pm	Zoom	TBA

Workshop	Pre-requisite*	Dates	Time	Location	Registration Page**
Introduction to Spatial Transcriptomics	Introduction to R	May 5, 8, 12 & 15, 2026	9:30am-12pm	Zoom	TBA
Introduction to scRNA-seq (Python-based)	Introduction to Python	August 11, 14, 18 & 21, 2026	9:30am-12pm	Zoom	TBA
Introduction to Spatial Transcriptomics	Introduction to R	October 6, 9, 13 & 16, 2026	9:30am-12pm	Zoom	TBA
Tools for Reproducible Research	Introduction to R	October 27, 30 & November 3, 2026	9:30am-12pm	Zoom	Register here

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